Week 4 Discussion

CSE 11 Winter 2026

Interfaces

- **Interface:** *a shared boundary, point of contact, or means of communication where two or more separate systems, devices, people, or fields meet and interact*
- Think of an interface as a **rule book**
  - Every record that implements an interface should have certain methods
  - Examples:
    - Animal interface, all animals records must have a speak method
    - Pict library, all Pict objects have a beside method
- **Cannot** directly create an instance

Interfaces

```
interface Animal {
    void speak();
}
```

```
record Dog implements Animal {
    void speak() {
        System.out.println("bark");
    }
}
```

```
record Cat implements Animal {
    void speak() {
        System.out.println("meow");
    }
}
```

```
Animal a = new Dog();
Animal b = new Cat();

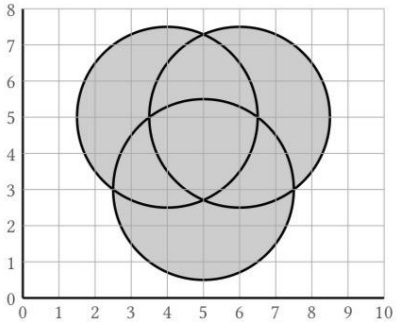
a.speak();   // woof
b.speak();   // meow
```

Memory diagram of the Union problem

- Not all objects in memory (the heap) need a reference pointing to it!
- On the worksheet yesterday, we saw that CircRegion implements a Region interface

```
Region one = new CircRegion(new Point(...), 2.5);
Region two = new CircRegion(new Point(...), 2.5);
Region three = new CircRegion(new Point(...), 2.5);

Region u1 = new Union(one, two);
Region all = new Union(u1, three);
```

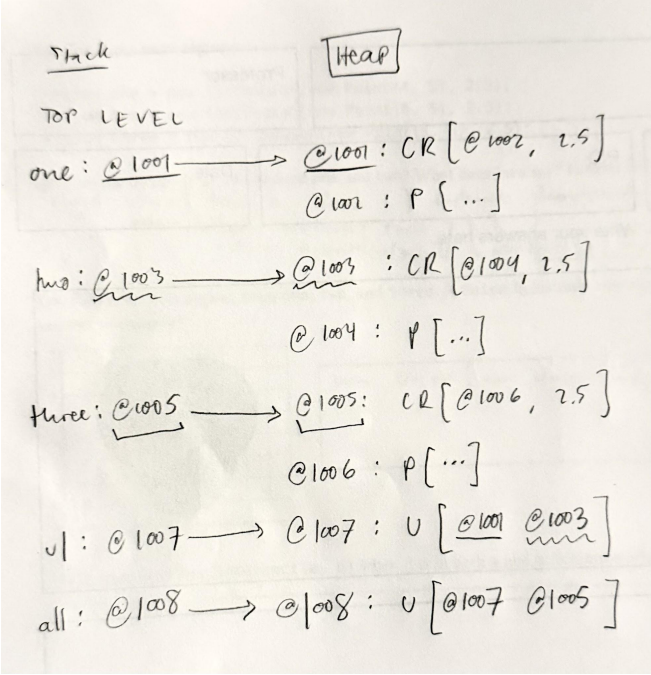


Question: Just by looking at this code, how many objects are created on the heap? How many variables are assigned on the stack?

Memory diagram of the Union problem

```
Region one = new CircRegion(new Point(...), 2.5);
Region two = new CircRegion(new Point(...), 2.5);
Region three = new CircRegion(new Point(...), 2.5);

Region u1 = new Union(one, two);
Region all = new Union(u1, all);
```



PA2 Overview

- Animations
  - Design and implement a small animation in Java using the Pict library and an array of frames `Pict[n]`
  - **Output**: `animation.gif` of your animation using `ImageIO_.save(...)` in Pict.java

Question 1:

- Write a program that draws a **circle** of radius 20 and color of your choice.
- Then draw a **rectangle** of height and width of 40 and color of your choice (so that the shapes are roughly the same size).
- Then save those objects in an array of frames.
- Then save the array of frames as an animation

```
java Animation.java 🚫
javac Animation.java && java Animation ✅
```

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```
1   public static void main(){
2
3       Pict c = new Circle(20, "blue");
4       Pict r = new Rect(40, 40, "red");
5       Pict[] frames = {c, r};
6
7       ImageIO_.save(frames, "animation.gif");
8   }
9
```

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Question 2:

- Modify your code to take in a single command line argument for the shape dimensions instead (i.e. for the radius of the circle and ½ of the rectangle's height and width)

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Example terminal commands:

`javac *.java && java Example2 50`

```
1   public static void main(String args[]){
2
3       int rad = Integer.parseInt(args[0]);
4
5       Pict c = new Circle(rad, "blue");
6       Pict r = new Rect(rad*2, rad*2, "red");
7       Pict[] frames = {c, r};
8
9       ImageIO_.save(frames, "animation2.gif");
10  }
```

PA2 Overview

- The solutions to all of these problems live in the animation workspace

Some Useful Things from the Pict Library

In the **`Raster`** class

- **`int width()`** – the length of each row
- **`int height()`** – the number of rows
- **`Color colorAt(int x, int y)`** – ***gets*** the pixel in row y, column x
- **`void setColorAt(int x, int y, Color color)`** – ***sets*** the pixel in row y, column x

Nested for-loops & Traversing 2D Arrays

```
static void traverse(int[][] pixels) {
    for (int i = 0; i < pixels.length; i++) {
        for (int j = 0; j < pixels[0].length; j++) {
            System.out.print(pixels[i][j] + " ");
        }
        System.out.println();
    }
}
```

What methods could you use inside the nested loops to **get and modify colors of a Raster object** instead?

Example of Using Raster from the Pict Library

```
import java.awt.Color;
Color averageColor(Raster raster) {
  int r = 0, g = 0, b = 0;
  int height = raster.height();
  int width = raster.width();
  for(int rowIndex = 0; rowIndex < height; rowIndex += 1) {
    for(int colIndex = 0; colIndex < width; colIndex += 1) {
      r += raster.colorAt(colIndex, rowIndex).getRed();
      g += raster.colorAt(colIndex, rowIndex).getGreen();
      b += raster.colorAt(colIndex, rowIndex).getBlue();
    }
  }
  int totalPix = width * height;
  int rAvg = r / totalPix, gAvg = g / totalPix, bAvg = b / totalPix;
  return new Color(rAvg, gAvg, bAvg);
}
Raster salk = ImageIO_.load("images/salk.png");
Color salkAvg = averageColor(salk);
```

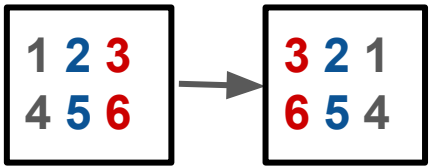
PA2 Overview

Put your answer to this question in the Gradescope submission!

- Filters
  - **Given**: a Photo object
  - **Return:** chainable methods that return new Photo values

Question 3:

- **Define a 2D array of integers.**
- **Use a nested for loop to flip this matrix of integers along the y-axis.**
- **Example:**

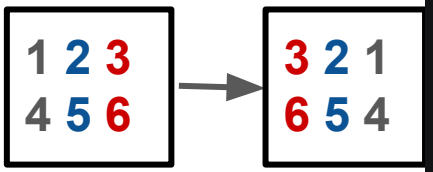


PA2 Overview

- Filters
  - **Given**: a Photo object
  - **Return:** chainable metho

Question 3:

- **Define a 2D array of intege**
- **Use a nested for loop to fli matrix of integers along th**
- **Example:**



Put your answer to this

Example3.java

```java
public static void main(String[] args){

    int[][] matrix = {
        {1, 2, 3},
        {4, 5, 6}
    };

    int nRows = matrix.length;
    int nCols = matrix[0].length;
    int[][] newMatrix = new int[nRows][nCols];
    for (int i = 0; i < nRows ; i++) {
        for (int j = 0; j < nCols ; j++) {
            System.out.println(matrix[i][j]);
            newMatrix[i][nCols - 1 - j] =  matrix[i][j];
        }
    }
    System.out.println(Arrays.deepToString(newMatrix));
}
```

PA2 Overview



- Barcode
  - **Given**: a serial number
  - **Return:** a picture of a barcode

Question 3:

- **Use command line arguments to write a program that prints a different message according to an int that you pass to your program.**
- **Print the following messages according to which integer the program is fed:**

0	"nope, not today"
1	"sure, why not?"
2	"maybe, what's the worst that could happen?"

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0	"nope, not today"
1	"sure, why not?"
2	"maybe, what's the worst that could happen?"

Example4.java

```java
public static void main(String[] args) {

    String[] responses = {
        "nope, not today",
        "sure, why not?",
        "maybe, what's the worst that could happen?"
    };

    int i = Integer.parseInt(args[0]);
    System.out.println(responses[i]);
}
```

TODO

- **PA02 is due THIS FRIDAY Oct 23**

Thank You!
Any Questions?